

Opt-in derived layer — never at ingest or crossmatch

Recipes
 per-survey JSON
 pack phot_ab_v1
 Vega→AB, flux→AB
 versioned + lineage

Homogenized product
 dl-homogenize
 products/<name>/
 AB (or nJy) columns
 subtype: homogenized

Science use
 dl-extract-*
 DuckDB / Polars
 ML training set
 FITS export

Geometry only — positional match, then join; photometry stays native

Area / region
 areas/<id>.json
 cone, bbox, MOC, npix
 metadata only
 optional plans

Crossmatch trees
 dl-crossmatch
 A_x_B_r<R>
 _source_id pairs + sep.
 reuse / gap-fill

Gathered product
 dl-gather
 products/<name>/
 wide native table
 kind: product

Native ingest — archival truth: no unit conversion

Catalogs
 catalogs/<survey>/
 HATS Parquet tiles
 Norder/Npix
 native columns & IDs

Spectra
 spectra/<survey>/
 sharded Zarr v3
 flux / ivar / mask
 shared or per-source λ

Cutouts
 cutouts/<survey>/
 sharded Zarr v3
 (N, B, H, W) float32
 per-source WCS

Housekeeping _source_id join key · _spectrum_npix/_cutout_npix · *_info.json
 lake registry & tile index · finalize / validate / rebuild indices

Ingest interchange: FITS / VOTable / CSV / Parquet →
 dl-ingest-catalog* / dl-ingest-spectra* / dl-ingest-cutouts*

path 2 (from product)

sky coords

path 1 (survey region)